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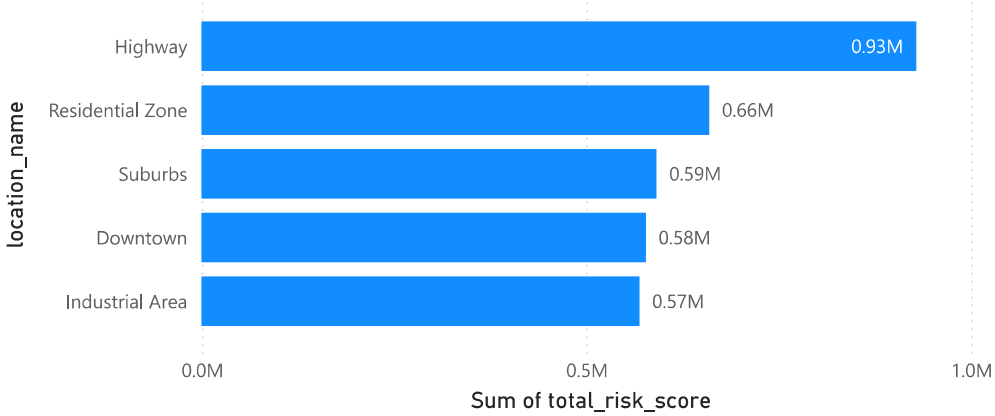
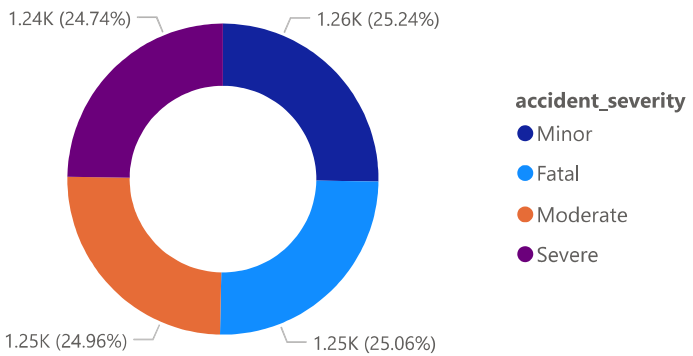
City Overview

5K	22K	1.253K	49.80%	11.04	0.49
Count of accident_id	Sum of casualties	Count of accident_severity	Fatal or Severe %	Avg Risk Score	Average of congestion_index

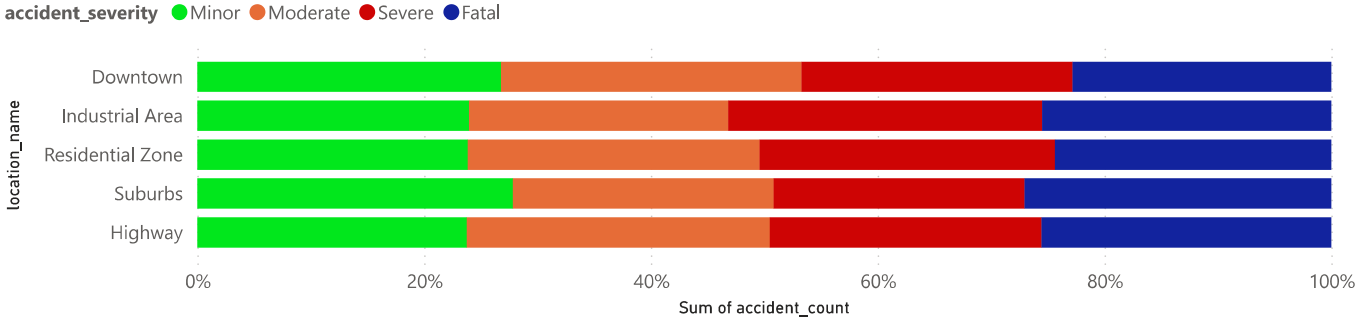
Total Risk Score by Location



Accident Severity Breakdown



Severity Profile by Location



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Traffic & Congestion

84K
Sum of vehicle_count

49.10
Average of...

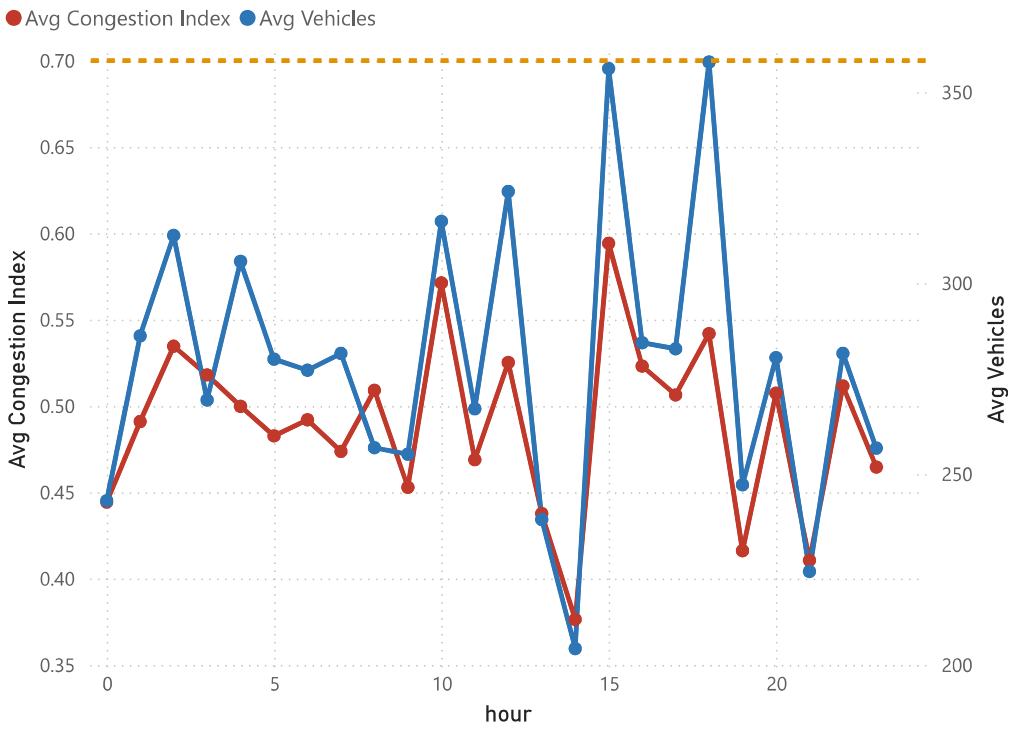
32.0%
High Congestion %

494
Max of vehicle_count

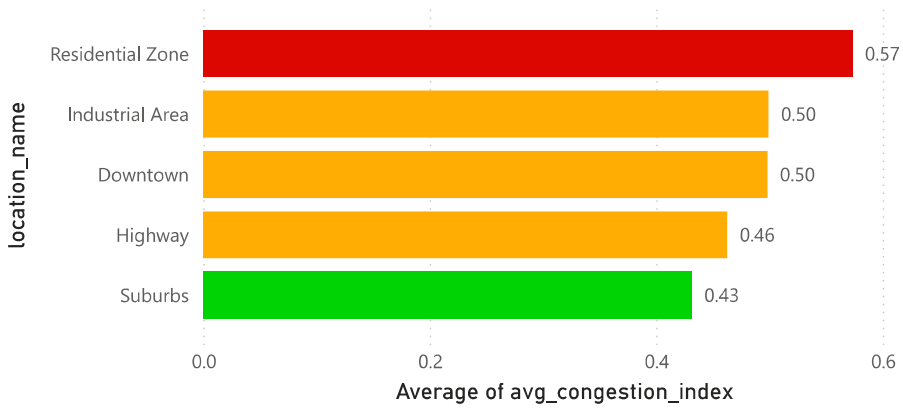
300
Sum of active_sensors

time_of_day
- ☐ Evening Peak
- ☐ Midday
- ☐ Morning Peak
- ☐ Off-Peak Day
- ☐ Off-Peak Night

24-Hour Traffic Profile - Volume & Congestion Index



Avg Congestion Index by Location



High Congestion % - Location * Time of Day

location_name	Evening Peak	Midday	Morning Peak	Off-Peak Day	Off-Peak Night	**Total**
Downtown	40.00	42.90	18.20	37.50	31.60	**170.20**
Highway	50.00	38.50	33.30	25.00	28.60	**175.40**
Industrial Area	12.50	20.00	30.00	36.40	23.50	**122.40**
Residential Zone	15.40	42.90	11.10	8.30	38.90	**116.60**
Suburbs	33.30	40.00	40.00	0.00	64.00	**177.30**
Total	**151.20**	**184.30**	**132.60**	**107.20**	**186.60**	**761.90**

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Accident Analysis

58.5%
Adverse Weather %

50.2%
Hazardous Road %

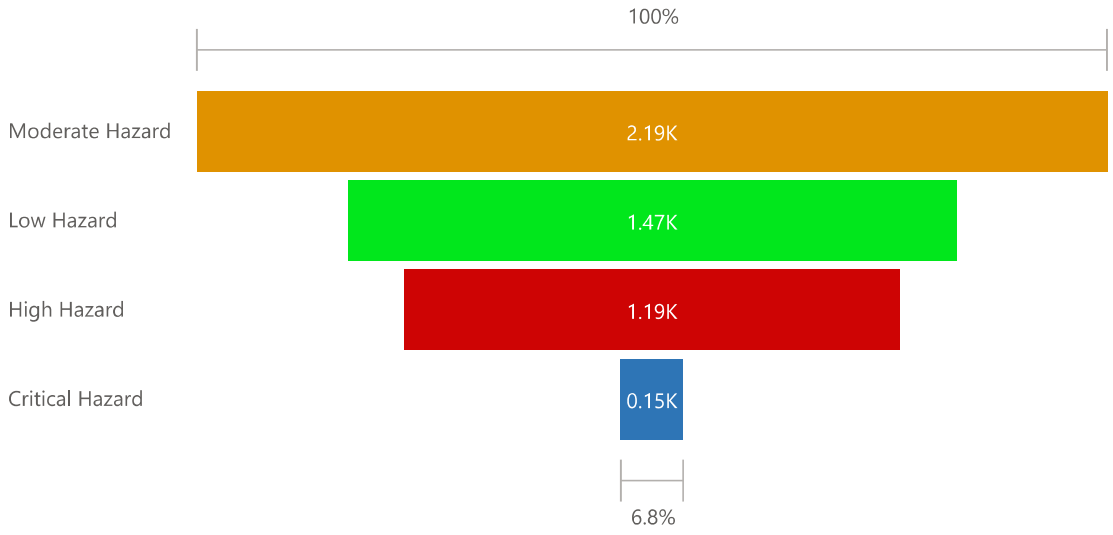
33.3%
Peak Hour Accidents %

4.49
Average of casualties

accident_severity
- [ ] Fatal
- [ ] Minor
- [ ] Moderate
- [ ] Severe

weather_condition
- [ ] Clear
- [ ] Fog
- [ ] Rain
- [ ] Snow
- [ ] Storm

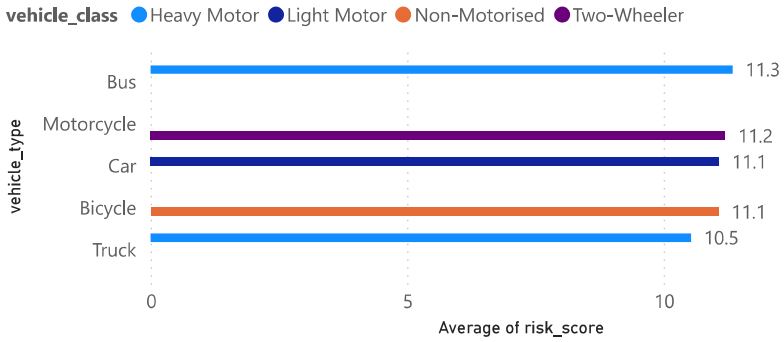
Accident Volume by Hazard Level



Avg Risk Score - Weather * Road Condition

weather_condition	Dry	Icy	Under Construction	Wet
Clear	12.22	11.50	10.48	11.21
Fog	10.86	10.56	10.81	10.79
Rain	11.72	11.04	11.21	10.97
Snow	11.29	10.84	10.94	11.17
Storm	10.85	11.21	11.65	9.36

Avg Risk Score by Vehicle Type



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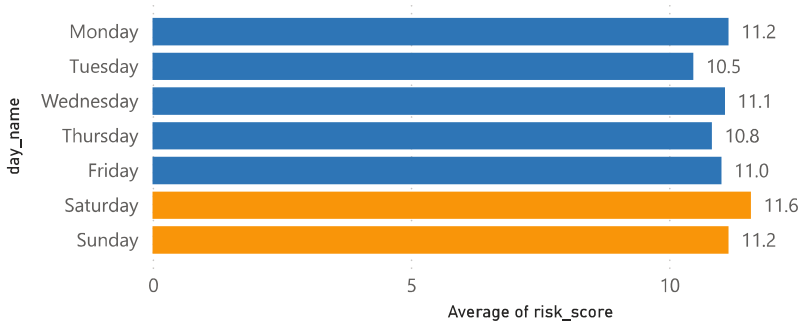
Time Trends

quarter

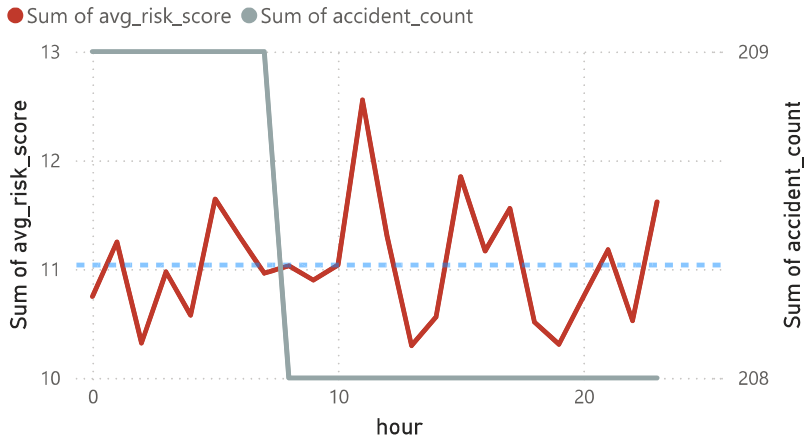
1 | 3

2

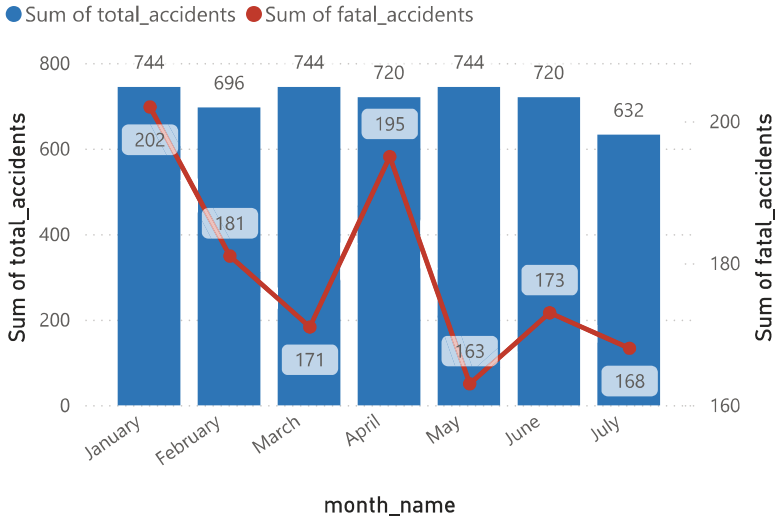
Accidents by Day of Week



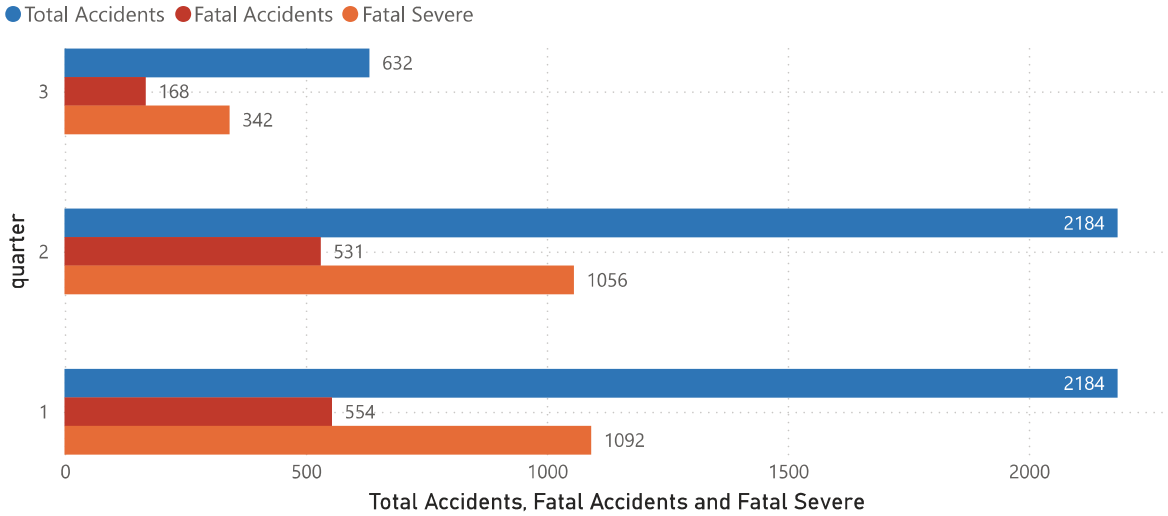
Hourly Accident Frequency bs Avg Risk Score



Monthly Accident & Fatality Trend (Jan-Jul 2024)



Quarterly Summary - Q1 vs Q2 (2024), Q3 = July only (partial data)



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Predictive Risk

safety_rating: Caution, Dangerous

risk_tier: Critical Risk, Low Risk, High Risk, Moderate Risk

0.50 Avg Predicted Risk

297 Critical Risk Count

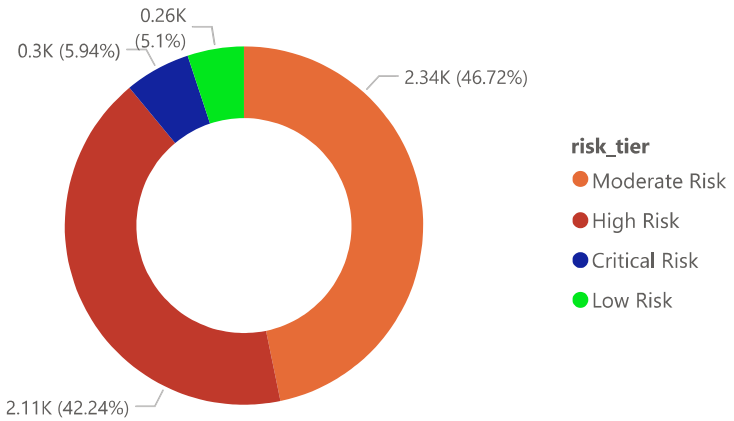
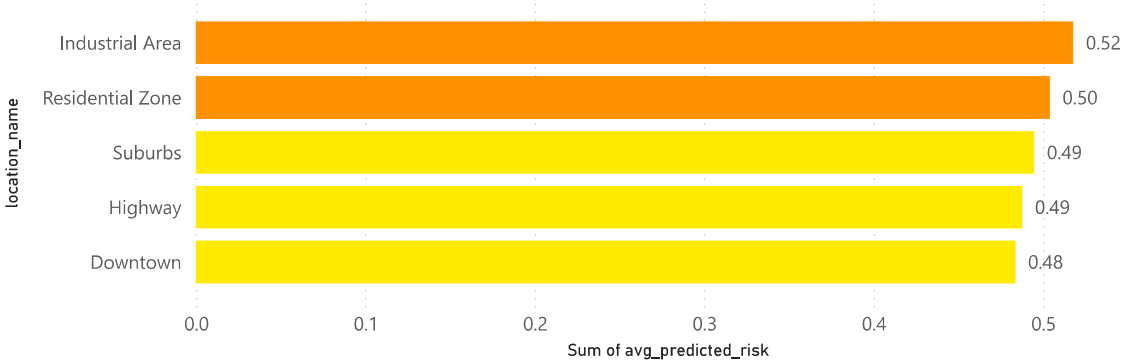
2K Predicted Fatal Severe Count

49.80 Actual Fatal Severe Rate

Location Safety Scorecard - ML Predictions vs Actual

Location Name	Avg Risk Prob	Safety Rating	Critical Risk #	Predicted Fatal/Severe	Actual Fatal/Severe %
Industrial Area	0.52	Dangerous	54.00	539.00	53.20
Residential Zone	0.50	Dangerous	64.00	482.00	50.40
Suburbs	0.49	Caution	63.00	493.00	49.20
Highway	0.49	Caution	53.00	451.00	49.50
Downtown	0.48	Caution	63.00	444.00	46.70
Total	**2.49**		**297.00**	**2,409.00**	**249.00**

ML Predicted Risk Tier Distribution(5k Accidents)



ML Predicted Risk by Location